

《密度泛函理论及应用》习题 1

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1 阅读 “A half century of density functional theory” 文章，翻译几段你认为有意义的或重要的话

In 1929 Paul Dirac made a famous announcement: The general theory of quantum mechanics is now almost complete.... The underlying physical laws necessary for the mathematical theory of a large part of physics and the whole of chemistry are thus completely known.... It therefore becomes desirable that approximate practical methods of applying quantum mechanics should be developed, which can lead to an explanation of the main features of complex atomic systems without too much computation.’

1929 年，保罗·狄拉克发表了一篇著名的声明：“量子力学的一般理论现在几乎已经完成……物理学的大部分和整个化学所需的基本物理定律因此被完全了解……因此，有必要开发近似的实用方法来应用量子力学，这些方法可以在不进行过多计算的情况下解释复杂原子系统的主要特征。”

It is important to appreciate the revolutionary nature of that question. In the Schrödinger equation, the ionic potential $v_{ext}(r)$ is the only term that distinguishes one alloy from another. That potential determines the wavefunction, which in turn determines the electron density and the total energy. The energy is thus a functional of $v_{ext}(r)$. Kohn contemplated a radical inversion of that thinking. Is it possible that the total energy depends only on the electron density $n(r)$? Years earlier Llewellyn Thomas and Enrico Fermi had proposed approximate theories of the total energy where that was the case, but no one had seriously suggested that the exact total energy of any many-electron system could be a functional of the electron density alone. If it were true, knowledge of $n(r)$ was sufficient to determine the external potential, the many-particle wavefunction, and all the ground-state properties—even the Green functions of many-body theory! That was a very deep question. Kohn realized he wasn’t doing alloy theory anymore.

理解这个问题的革命性本质是很重要的。在薛定谔方程中，离子势 $v_{ext}(r)$ 是区分一种合金与另一种合金的唯一项。该势决定了波函数，而波函数又决定了电子密度和总能量。因此，总能量是 $v_{ext}(r)$ 的泛函。科恩考虑了一种彻底颠倒这种思维方式的可能性。总能量是否仅依赖于电子密度 $n(r)$? 多年前，Llewellyn Thomas 和 Enrico Fermi 提出了总能量的近似理论，但没有人认真建议任何多电子系统的精确总能量可以仅仅是电子密度的泛函。如果这是真的，那么对 $n(r)$ 的了解就足以确定外部势、多粒子波函数以及所有基态性质——甚至是多体理论的格林函数！这是一个非常深刻的问题。科恩意识到他不再是在做合金理论了。

General questions require general solution methods, and Kohn was particularly well trained in one class of them: the variational methods of mathematical physics. Such methods were a particular specialty of the department of applied mathematics at the University of Toronto, where Kohn had earned his BA and MA degrees. The local experts there were chairperson John Lighton Synge and two of Kohn's mentors, Alexander Weinstein and Arthur Stevenson. Another virtuoso of variational methods, Julian Schwinger, supervised his Harvard PhD thesis, which introduces what is today called the Kohn variational principle. From the reminiscence paper, or from any technical review of DFT, one can learn how variational methods are central to the proof of the Hohenberg–Kohn theorem and to the derivation of the Kohn–Sham equations.

一般性的问题需要一般性的解决方法，而科恩在其中一类方法上受过特别好的训练：数学物理的变分方法。这些方法是多伦多大学应用数学系的一个特殊专业，科恩在那里获得了他的学士和硕士学位。当地的专家有系主任 John Lighton Synge 和科恩的两位导师 Alexander Weinstein 和 Arthur Stevenson。另一位变分方法的大师 Julian Schwinger 监督了他的哈佛博士论文，该论文引入了今天称为 Kohn 变分原理的方法。从回忆录论文或任何关于 DFT 的技术评论中，人们可以了解到变分方法在 Hohenberg-Kohn 定理的证明和 Kohn-Sham 方程的推导中是多么重要。

2 写出静电场下单个氢气分子的精确哈密顿量 (使用原子单位)

$$\begin{aligned}
 \hat{H} &= \hat{T}_n + \hat{T}_e + V_{nn} + V_{ee} + V_{en} + V_{ext} \\
 &= \sum_{\alpha} -\frac{1}{2M_{\alpha}} \nabla_{\alpha}^2 - \sum_i \frac{1}{2} \nabla_i^2 + \sum_{\alpha < \beta} \frac{Z_{\alpha} Z_{\beta}}{|R_{\alpha} - R_{\beta}|} + \sum_{i < j} \frac{1}{|r_i - r_j|} + \sum_{\alpha, i} \frac{-Z_{\alpha}}{|R_{\alpha} - r_i|} - E \cdot \mu \\
 &= -\frac{1}{2M_p} (\nabla_{R_1}^2 + \nabla_{R_2}^2) - \frac{1}{2} (\nabla_{r_1}^2 + \nabla_{r_2}^2) + \frac{1}{|R_1 - R_2|} + \frac{1}{|r_1 - r_2|} \\
 &\quad - \left(\frac{1}{|R_1 - r_1|} + \frac{1}{|R_1 - r_2|} + \frac{1}{|R_2 - r_1|} + \frac{1}{|R_2 - r_2|} \right) + E(r_1 + r_2) - E(R_1 + R_2)
 \end{aligned}$$

注：此处的 E 为外电场强度。

3 把上一题中的哈密顿量，利用绝热近似分成原子核部分和电子部分

电子部分

$$\begin{aligned}
 \hat{H}_{el} &= \hat{T}_e + \hat{U}_{ee} + V_{en} + V_{ext} \\
 &= -\frac{1}{2} (\nabla_{r_1}^2 + \nabla_{r_2}^2) - \sum_{i=1}^2 \sum_{j=1}^2 \frac{1}{|R_i - r_j|} + \frac{1}{|r_1 - r_2|} + E(r_1 + r_2)
 \end{aligned}$$

原子核部分

$$\begin{aligned}\hat{H}_{nl} &= \hat{T}_n + \hat{V}_{nn} + \hat{V}_{ext} \\ &= -\frac{1}{2M_p}(\nabla_{R_1}^2 + \nabla_{R_2}^2) + \frac{1}{|R_1 - R_2|} - E(R_1 + R_2)\end{aligned}$$

4 在 Hartree-Fock 近似下, 推导 $\langle \Psi(r_1, r_2) | \frac{1}{|r_1 - r_2|} | \Psi(r_1, r_2) \rangle$ 的表达式

$$\begin{aligned}&\langle \Psi(r_1, r_2) | \frac{1}{|r_1 - r_2|} | \Psi(r_1, r_2) \rangle \\ &= \frac{1}{2} \int (\phi_1(r_1)\phi_2(r_2) - \phi_2(r_1)\phi_1(r_2))^* \frac{1}{|r_1 - r_2|} (\phi_1(r_1)\phi_2(r_2) - \phi_2(r_1)\phi_1(r_2)) dr_1 dr_2 \\ &= \frac{1}{2} \int \phi_1^*(r_1)\phi_2^*(r_2) \frac{1}{|r_1 - r_2|} \phi_1(r_1)\phi_2(r_2) dr_1 dr_2 - \frac{1}{2} \int \phi_1^*(r_1)\phi_2^*(r_2) \frac{1}{|r_1 - r_2|} \phi_2(r_1)\phi_1(r_2) dr_1 dr_2 \\ &\quad - \frac{1}{2} \int \phi_2^*(r_1)\phi_1^*(r_2) \frac{1}{|r_1 - r_2|} \phi_1(r_1)\phi_2(r_2) dr_1 dr_2 + \frac{1}{2} \int \phi_2^*(r_1)\phi_1^*(r_2) \frac{1}{|r_1 - r_2|} \phi_2(r_1)\phi_1(r_2) dr_1 dr_2 \\ &= \frac{1}{2} \int \phi_1^*(r)\phi_2^*(r') \frac{1}{|r - r'|} \phi_1(r)\phi_2(r') dr dr' - \frac{1}{2} \int \phi_1^*(r)\phi_2^*(r') \frac{1}{|r - r'|} \phi_2(r)\phi_1(r') dr dr' \\ &\quad - \frac{1}{2} \int \phi_2^*(r')\phi_1^*(r) \frac{1}{|r' - r|} \phi_1(r')\phi_2(r) dr dr' + \frac{1}{2} \int \phi_2^*(r')\phi_1^*(r) \frac{1}{|r' - r|} \phi_2(r')\phi_1(r) dr dr' \\ &= \int \frac{\phi_1^*(r)\phi_1(r)\phi_2^*(r')\phi_2(r')}{|r - r'|} dr dr' - \int \frac{\phi_1^*(r)\phi_1(r')\phi_2^*(r')\phi_2(r)}{|r - r'|} dr dr'\end{aligned}$$

5 在 Hartree-Fock 近似下, 说明关联能的来源及其定义, 说明关联能不会大于零, 并列举解决电子关联问题的常用方法

所谓的“关联”指波函数不能写为单粒子波函数的乘积形式:

$$\Phi(r_1, r_2, \dots, r_n) \neq \phi(r_1)\phi(r_2)\cdots\phi(r_n)$$

因为使用单电子波函数的乘积不是系统的基态, 计算出来的总能量比真实基态的总能量高, 它们之间差值就是关联能。

$$E_c = \langle H \rangle - \langle H \rangle_{HF} \leq 0$$

常用的修正电子的关联相互作用的方法有: Configuration Interaction (CI), Coupled Cluster (CC), Møller-Plesset Perturbation Theory ...

6 利用变分原理推导 Hartree-Fock 方法交换积分的变分导数

$$\begin{aligned}
\delta \left[\frac{1}{2} \sum_{ij} K_{ij} \right] &= \frac{1}{2} \delta \left[\sum_{ij} \iint \phi_i^*(r_1) \phi_j^*(r_2) \frac{1}{|r_1 - r_2|} \phi_i(r_2) \phi_j(r_1) dr_1 dr_2 \right] \\
&= \frac{1}{2} \sum_j \iint \delta \phi_a^*(r_1) \phi_j^*(r_2) \frac{1}{|r_1 - r_2|} \phi_a(r_2) \phi_j(r_1) dr_1 dr_2 \\
&\quad + \frac{1}{2} \sum_i \iint \delta \phi_i^*(r_1) \phi_a^*(r_2) \frac{1}{|r_1 - r_2|} \phi_i(r_2) \phi_a(r_1) dr_1 dr_2 \\
&= \sum_i \iint \delta \phi_a^*(r_1) \phi_i^*(r_2) \frac{1}{|r_1 - r_2|} \phi_a(r_2) \phi_i(r_1) dr_1 dr_2 \\
\\
\frac{\delta \left[\frac{1}{2} \sum_{ij} K_{ij} \right]}{\delta \phi_a^*(r_1)} &= \sum_i \iint \phi_i^*(r_2) \frac{1}{|r_1 - r_2|} \phi_a(r_2) \phi_i(r_1) dr_1 dr_2 \\
&= \sum_i \phi_i(r_1) \int \phi_i^*(r_2) \frac{1}{|r_1 - r_2|} \phi_a(r_2) dr_2 \\
&= \sum_i \phi_i(r_1) \int \frac{\phi_i^*(r_2) \phi_a(r_2)}{|r_1 - r_2|} dr_2
\end{aligned}$$

7 对比 Thomas-Fermi 理论和密度泛函理论，说明密度泛函理论在哪些方面做了改进

TFD 理论提出总能量是电子密度的泛函，但是其中电子动能项的密度泛函形式过于简单，导致计算误差较大。

DFT 理论不是使用电子密度，而是使用电子波函数来计算动能项。这种方法计算精度高，避免动能项没有精确泛函的问题，同时把动能项和电子间库仑相互作用的误差都放在交换关联势中，这部分能量占比较小。

8 写出 Kohn-Sham 方程，详细说明其中每一项物理意义

Kohn-Sham equation:

$$\begin{aligned}
\left[-\frac{1}{2} \nabla^2 + \hat{V}_{ext} + \hat{U}_{cl} + \hat{V}_{xc}(r) \right] \phi_i^{KS}(r) &= \varepsilon_i \phi_i^{KS}(r) \\
\left[-\frac{1}{2} \nabla^2 + \sum_{\alpha} -\frac{Z_{\alpha}}{|R_{\alpha} - r|} + \int \frac{\rho(r')}{|r - r'|} dr' + \frac{\delta E_{xc}[\rho(r)]}{\delta \rho(r)} \right] \phi_i^{KS}(r) &= \varepsilon_i \phi_i^{KS}(r)
\end{aligned}$$

其中，第一项 $-\frac{1}{2} \nabla^2$ 表示电子的动能算符，描述电子在空间中的运动行为。

第二项 $\hat{V}_{ext}$ 表示外势场算符，通常由原子核对电子的吸引作用构成，决定了电子在原子或分子中的分布。

第三项 $\hat{U}_{cl}$ 表示经典库伦势算符，描述电子之间的静电相互作用，即电子间的排斥效应。

最后一项 $\hat{V}_{xc}(r)$ 表示交换-关联势算符，包含了电子间的交换和关联效应，是密度泛函理论中最复杂且关键的部分。

右侧的 ε_i 是 Kohn-Sham 轨道的能量本征值, $\phi_i^{KS}(r)$ 是对应的 Kohn-Sham 轨道波函数。

9 当前流行的交换关联函主要有哪些？试比较它们的各自应用特点

- LDA
共价系统, 简单金属
- DFT+U
Mott 绝缘体, 强关联材料
- GGA
分子, 氢键材料, 密度变化大的 d 态及 f 态, 大多数磁性材料, 特殊的金属
- Non-local hybrids and sX
带隙, 特殊的磁性材料
- van der Waals
层状材料等

10 以氢原子为例, 根据量子力学求解出来的波函数, 计算其在 LDA 下的动能、外势场能、库伦能、交换能和关联能 (PW92)

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}$$

$$dr = r^2 \sin \theta dr d\theta d\phi$$

$$\Phi = \frac{1}{\sqrt{\pi}} e^{-r}$$

$$\rho = \Phi^* \Phi = \frac{1}{\pi} e^{-2r}$$

$$\begin{aligned} T &= \int \Phi^* \left(-\frac{1}{2} \nabla^2 \right) \Phi dr = -\frac{1}{2\pi} \int e^{-r} \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) e^{-r} \right] dr \\ &= \frac{1}{2\pi} \int e^{-r} \frac{2r - r^2}{r^2} e^{-r} r^2 dr \sin \theta d\theta d\phi = 2 \int (2r - r^2) e^{-2r} dr = \frac{1}{2} \end{aligned}$$

$$E_{ext} = \int \Phi^* \left(-\frac{1}{r} \right) \Phi dr = -\frac{1}{\pi} \int \frac{1}{r} e^{-r} r^2 dr \sin \theta d\theta d\phi = -4 \int r e^{-2r} dr = -1$$

$$\begin{aligned} E_x &= \int \rho \varepsilon_x(\rho) dr = -\frac{3}{4} \left(\frac{3}{\pi} \right)^{\frac{1}{3}} \int \rho^{\frac{4}{3}} r^2 dr \sin \theta d\theta d\phi \\ &= -3 \left(\frac{3}{\pi^2} \right)^{\frac{1}{3}} \int r^2 e^{-\frac{8}{3}r} dr = -\left(\frac{3}{4} \right)^4 \left(\frac{3}{\pi^2} \right)^{\frac{1}{3}} = -0.2127 \end{aligned}$$

$$\rho \varepsilon_c(\rho) = \rho(-2A)(1 + \alpha_1 r_s) \ln \left(1 + \frac{1}{2A(\beta_1 r_s^{\frac{1}{2}} + \beta_2 r_s + \beta_3 r_s^{\frac{3}{2}} + \beta_4 r_s^2)} \right)$$

$$r_s = \left(\frac{3}{4\pi\rho} \right)^{1/3}, \quad \rho = \frac{1}{\pi} e^{-2r}$$

$$A = 0.031091, \quad \alpha_1 = 0.2137, \quad \beta_1 = 7.5957, \quad \beta_2 = 3.5876, \quad \beta_3 = 1.6382, \quad \beta_4 = 0.49294$$

$$E_c = \int \rho \varepsilon_c(\rho) dr = \int \rho \varepsilon_c(\rho) r^2 dr \sin \theta d\theta d\phi = 4\pi \int \rho \varepsilon_c(\rho) r^2 dr = -0.0414$$

$$\begin{aligned} E_{cl} &= \frac{1}{2} \iint \frac{\rho(r)\rho(r')}{|r-r'|} dr dr' = \frac{1}{2} \int dr \rho(r) \int \frac{\rho(r')}{|r-r'|} dr' \\ &= \frac{1}{2} \int dr \rho(r) U(r) \end{aligned}$$

$$\begin{aligned} U(r) &= \int \frac{\rho(r')}{|r-r'|} dr' \\ &= \frac{1}{\pi} \int \frac{e^{-2r'}}{\sqrt{r^2 + r'^2 - 2rr' \cos \theta'}} r'^2 \sin \theta' dr' d\theta' d\phi' \\ &= 2 \int dr' r'^2 e^{-2r'} \int \frac{\sin \theta' d\theta'}{\sqrt{r^2 + r'^2 - 2rr' \cos \theta'}} \\ &= 2 \int dr' r'^2 e^{-2r'} \frac{1}{2rr'} \int \frac{d(r^2 + r'^2 - 2rr' \cos \theta')}{\sqrt{r^2 + r'^2 - 2rr' \cos \theta'}} \\ &= \int dr' r'^2 e^{-2r'} \frac{2}{rr'} \sqrt{r^2 + r'^2 - 2rr' \cos \theta'} \Big|_0^\pi \\ &= \int dr' r'^2 e^{-2r'} \frac{2}{rr'} (r + r' - |r - r'|) \end{aligned}$$

$$U(r) = \begin{cases} \frac{4}{r} \int dr' r'^2 e^{-2r'}, & r > r' \\ 4 \int dr' r' e^{-2r'}, & r \leq r' \end{cases} = \frac{1}{r} - \frac{1}{r} e^{-2r} - e^{-2r}$$

$$\begin{aligned} E_{cl} &= \frac{1}{2} \iint \frac{\rho(r)\rho(r')}{|r-r'|} dr dr' = \frac{1}{2} \int dr \rho(r) U(r) \\ &= \frac{1}{2} \int dr \rho(r) \left(\frac{1}{r} - \frac{1}{r} e^{-2r} - e^{-2r} \right) \\ &= \frac{1}{2\pi} \int e^{-2r} \left(\frac{1}{r} - \frac{1}{r} e^{-2r} - e^{-2r} \right) r^2 dr \sin \theta d\theta d\phi \\ &= 2 \int (r e^{-2r} - r e^{-4r} - r^2 e^{-4r}) dr \\ &= \frac{5}{16} \end{aligned}$$